



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

FULL SYLLABUS

11-05-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE/ALL STAR BATCH/(SCORE ADVANCED)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	A	C	B	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	B,C	A,B,C	A,D	A,B,C	A,B,C	A,B,C		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	2	8	0	8	2	9	2	2

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4				
	A.	D	C	B	D				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A,C,D	A,B	A,B,C,D	B,C,D	A,B,D	A,C		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	3	4	4	8	7	4	6	8

PART-3 : MATHEMATICS

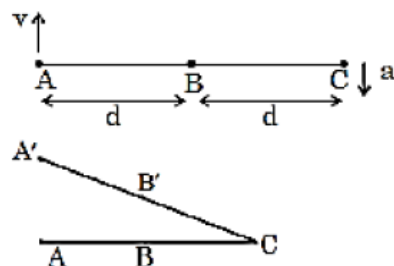
SECTION-I (i)	Q.	1	2	3	4				
	A.	D	B	D	D				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A,C	A,B,C	B,C	B,D	B,C,D	A,C,D		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	3	4	4	3	0	1	8	7

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A)



Let us observe the motion of A and B relative to C.

$$\frac{AA'}{2d} = \frac{BB'}{d}$$

$$BB' = \frac{AA'}{2} = \frac{vt + \frac{1}{2}at^2}{2}$$

$$BB' = \frac{Vt}{2} + \frac{1}{4}at^2$$

$\frac{V}{2}$ is initially speed of B w.r.t C as well as ground.

$\frac{a}{2}$ is acceleration of B w.r.t C

$$\vec{a}_{B/g} = \vec{a}_{B/C} + \vec{a}_{C/g}$$

$$= \frac{a}{2} - a = -\frac{1}{2}a$$

4. Ans (B)

The wavelength at which the intensity of radiation is maximum depends only on the temperature. As this wavelength is the same for both, their temperatures must be equal. The difference in intensities can then be due only to a difference in their emissivities.

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (B,C)

Negative zero error = -1.25

Thickness = $18 + 0.34 - (-1.25) = 19.59 \text{ mm}$

6. Ans (A,B,C)

Speed = $v = \sqrt{\frac{T}{\mu}} = 50 \text{ m/s}$

Power = $\frac{1}{2} \mu \omega^2 a^2 v = 0.126 \text{ W}$

$E_{\text{avg}} = P_{\text{avg}} \cdot t = P_{\text{avg}} \left(\frac{1}{9} \right) = 2.5 \text{ mJ}$

7. Ans (A,D)

$$\begin{aligned}
 U &= \int_R^{2R} \frac{1}{2} \epsilon_0 E^2 dV = \frac{1}{2} \epsilon_0 \frac{K^2 Q^2}{r^4} 4\pi r^2 dr \\
 &= \frac{KQ^2}{2} \int_R^{2R} r^{-2} dr = \frac{Q^2}{8\pi\epsilon_0} \left[\frac{1}{-2R} + \frac{1}{R} \right] \\
 &= \frac{Q^2}{16\pi\epsilon_0 R}
 \end{aligned}$$

Now no electric field inside sphere.

So $U = 0$ (inside)

$$U_{\text{self}} = \frac{KQ^2}{2R} = \text{heat lost}$$

8. Ans (A,B,C)

Potential difference is defined only for electrostatic field.

9. Ans (A,B,C)

$$\begin{aligned}
 \frac{n_p}{n_p \left(\frac{d}{n_p} + \frac{t}{1} + \frac{x}{n_g} \right)} + \frac{1}{\infty} &= \frac{(1 - n_p)}{-R} \\
 \Rightarrow \frac{d}{n_p} + t &= \frac{R}{n_p - 1} - \frac{x}{n_g}
 \end{aligned}$$

10. Ans (A,B,C)

$$Q = \frac{h^2}{24m\lambda^2} = K_B + K_C$$

Where k is the kinetic energy

From conservation of linear momentum

$$|\vec{P}_C| = |\vec{P}_A| = P$$

$$\therefore K = \frac{P^2}{2m}$$

$$\begin{aligned}
 \therefore Q &= \frac{P^2}{2 \times 12m} + \frac{P^2}{2 \times 4m} \\
 \frac{h^2}{24m\lambda^2} &= \frac{P^2}{2} \left[\frac{1}{12m} + \frac{1}{4m} \right] \\
 \frac{h^2}{12m\lambda^2} &= P^2 \left(\frac{1+3}{12m} \right) = \frac{P^2}{3m}
 \end{aligned}$$

$$\Rightarrow P^2 = \frac{h^2 \times 3m}{12m\lambda^2} \Rightarrow P = \frac{h}{2\lambda}$$

$$\Rightarrow \frac{h}{P} = 2\lambda = \text{de-Broglie wavelength}$$

$$\therefore |\vec{P}_B| = |\vec{P}_C|$$

$$\text{Mass defect, } \Delta m = \frac{Q}{C^2} = \frac{h^2}{24mc^2\lambda^2}$$

$$\therefore m_A = 16m + \frac{h^2}{24mc^2\lambda^2}$$

PART-1 : PHYSICS

SECTION-III

1. Ans (2)

$$k = \frac{2\pi}{\lambda} = 62.8\lambda = 0.1 \text{ m} \Rightarrow \lambda = 2 \text{ m}$$

2. Ans (8)

$$Kx \cos 37^\circ = T \cos 53^\circ$$

$$Kx \cos 53^\circ + T \cos 37^\circ = mg$$

$$\Rightarrow \frac{T (\cos^2 53^\circ + \cos^2 37^\circ)}{\cos 37^\circ} = mg$$

$$\Rightarrow T = \frac{4mg}{5}$$

$$K\vec{x} + \vec{T} + m\vec{g} = 0$$

$K\vec{x}$ does not change suddenly

$$\Rightarrow K\vec{x} + m\vec{g} = m\vec{a} = -\vec{T}$$

$$ma = \frac{4mg}{5}$$

$$\Rightarrow a = 8 \text{ m/s}^2$$

3. Ans (0)

When slider is at A, the applied potential difference across the photoelectric cell is a retarding potential difference of 10 V. Energy of incident photon

$$E = \frac{hc}{\lambda} = \frac{1240 \text{ eV nm}}{350 \text{ nm}} = 3.5 \text{ eV}$$

Maximum KE of emitted photoelectrons is

$$KE_{\max} = \frac{hc}{\lambda} - \phi = 3.5 - 2.2 = 1.3 \text{ eV}$$

stopping potential difference

$$V_s = 1.3 \text{ V}$$

(a) The applied potential difference is

$$V (= 10 \text{ V}) > V_s (= 1.3 \text{ V})$$

Hence there is no photocurrent and reading of ammeter is zero.

4. Ans (8)

By the conservation of energy

$$\frac{1}{2} m V_0^2 = mgh + \frac{1}{2} m v^2 \dots \dots \dots (1)$$

By the conservation of angular momentum

$$\frac{m V_0 r}{2} = m V R \dots \dots \dots (2)$$

From (1) & (2)

$$V_0 = \sqrt{\frac{8gh}{3}}$$

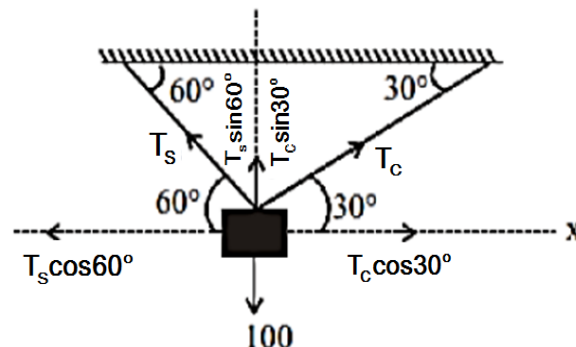
$$k = 8$$

5. Ans (2)

Let T_S = tension in steel wire

T_C = Tension in copper wire

In x direction



$$T_C \cos 30^\circ = T_S \cos 60^\circ$$

$$T_C \times \frac{\sqrt{3}}{2} = T_S \times \frac{1}{2}$$

$$\sqrt{3} T_C = T_S \dots \dots \dots (i)$$

In y direction

$$T_C \sin 30^\circ + T_S \sin 60^\circ = 100$$

$$\frac{T_C}{2} + \frac{T_S \sqrt{3}}{2} = 100 \dots \dots \dots (ii)$$

Solving equation (i) & (ii)

$$T_C = 50 \text{ N}$$

$$T_S = 50\sqrt{3} \text{ N}$$

$$\text{We know } \Delta L = \frac{FL}{AY}$$

$$= \frac{\Delta L_C}{\Delta L_S} = \frac{T_C L_C}{T_S L_S} \times \frac{A_S Y_S}{A_C Y_C}$$

On solving above equation

$$\frac{\Delta L_C}{\Delta L_S} = 2$$

6. Ans (9)

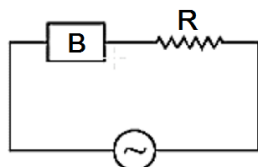
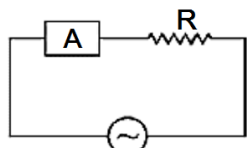
Since water resistance ($\vec{F} = -k\vec{v}_{pr}$) is proportional to the velocity of plank relative to the water, impulse of water resistance equals to product of the proportionality constant k and displacement of the plank relative to the water. And this impulse stops the block- plank system at the other bank, therefore equals to the initial momentum of the block.

$$m u_{\min} = k \int v dt$$

$$\Rightarrow m u_{\min} = k(b - l)$$

$$u_{\min} = \frac{k(b - l)}{m} = 9.0 \text{ m/s}$$

7. Ans (2)



$$\phi_1 = -30^\circ$$

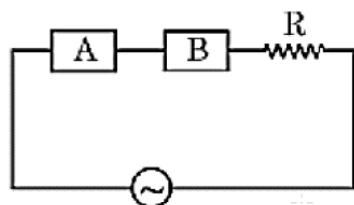
$$\phi_2 = +60^\circ$$

$$\tan \phi = \frac{X_L}{R}$$

$$\tan \phi_2 = \frac{X_C}{R}$$

$$X_L = \frac{R}{\sqrt{3}}$$

$$X_C = R\sqrt{3}$$



$$\tan \theta = \frac{(X_C - X_L)}{R} = \frac{\frac{R\sqrt{3}-R}{\sqrt{3}}}{R} = \frac{2}{\sqrt{3}}$$

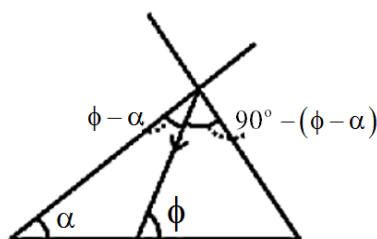
8. Ans (2)

$$a_1 \cos[90^\circ - (\phi - \alpha)] = a_2 \sin \alpha$$

$$a_1 \sin(\phi - \alpha) = a_2 \sin \alpha \dots\dots\dots (i)$$

Conservation has no acceleration x direction

$$\Rightarrow Ma_2 = ma_1 \sin \phi \dots\dots\dots (ii)$$



From (i) and (ii)

$$\Rightarrow \frac{m}{M} \sin \phi = \frac{\sin(\phi - \alpha)}{\sin \alpha}$$

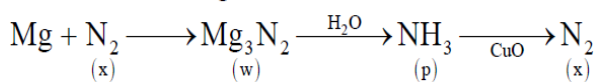
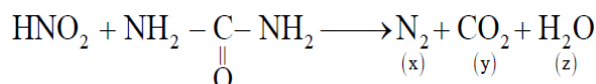
$$\Rightarrow \frac{m}{M} = \frac{\tan \phi - \tan \alpha}{\tan \alpha}$$

$$\frac{m}{M} = \frac{\tan 60^\circ - \tan 30^\circ}{\tan 30^\circ} = 2$$

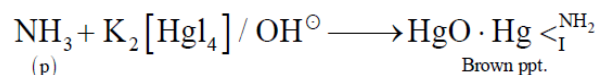
PART-2 : CHEMISTRY

SECTION-I (i)

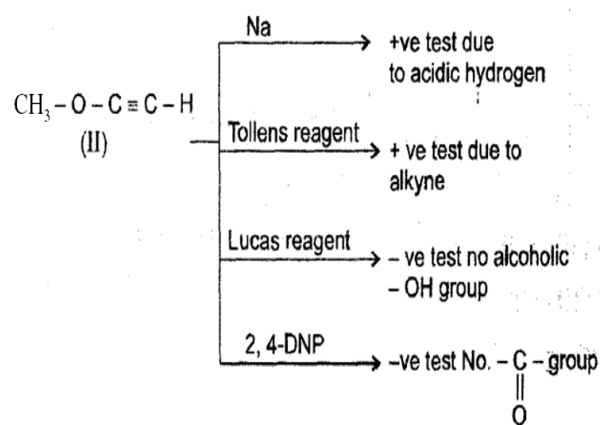
1. Ans (D)



Milky solution



2. Ans (C)



3. Ans (B)

From the equation,

$$[\text{Cu}(\text{CN})_4]^{3-} = 0.1\text{M} \text{ and } [\text{CN}^-] = 0.2\text{M}$$

$$\therefore [\text{Cu}^+] = \frac{K_{\text{instab.}} [\text{Cu}(\text{CN})_4]^{3-}}{[\text{CN}^-]^4}$$

$$= \frac{6.4 \times 10^{-15} \times 0.1}{(0.2)^4} = 4 \times 10^{-13}\text{M}$$

$$\text{Now, } [\text{S}^{2-}] = \frac{K_{\text{SP}}(\text{Cu}_2\text{S})}{[\text{Cu}^+]^2} = \frac{2.56 \times 10^{-27}}{(4 \times 10^{-13})^2}$$

$$= 1.6 \times 10^{-2}\text{M}$$

$$\therefore [\text{H}^+] = \sqrt{\frac{K_a \times [\text{H}_2\text{S}]}{[\text{S}^{2-}]}} = \sqrt{\frac{1.6 \times 10^{-21}}{1.6 \times 10^{-2}}}$$

$$= 10^{-10}\text{M} \text{ and } \text{pH} = 10.0$$

4. Ans (D)

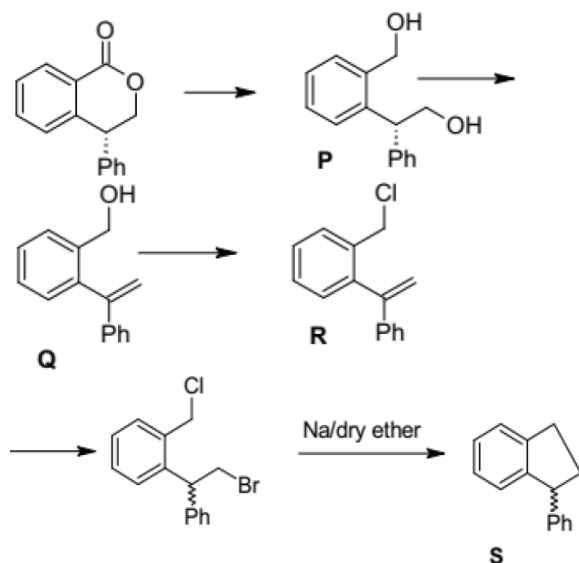
$$n = (1+4+9+\dots+n^2) = \frac{P_f(1+2+3+\dots+n)V}{RT}$$

$$P_f = \frac{P \left[\frac{n(n+1)(2n+1)}{6} \right]}{\frac{n(n+1)}{2}} = \left(\frac{2n+1}{3} \right) P$$

PART-2 : CHEMISTRY

SECTION-I (ii)

6. Ans (A,B)



7. Ans (A,B,C,D)

A) It is deep red colouration due to the formation of $\text{Fe}(\text{SCN})_3$.

B) In presence of moisture following reaction will occur:

$\text{CrO}_2\text{Cl}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{CrO}_4 + \text{HCl}$; So test is carried out only in dry test tube.

C) White precipitate of PbSO_4 is formed and hence brown ring is not visible.

D) Gives yellow precipitate of CdS .

8. Ans (B,C,D)

A) $2\text{HCO}_3^- + \text{Mg}^{2+} \xrightarrow{\text{cold}} \text{Mg}(\text{HCO}_3)_2$ (soluble)
 $\xrightarrow{\Delta}$

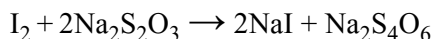
$\text{MgCO}_3 \downarrow + \text{CO}_2 + 2\text{H}_2\text{O}$

B) $\text{Ca}(\text{HCO}_3)_2 + 2\text{NH}_3 \rightarrow (\text{NH}_4)_2\text{CO}_3 + \text{CaCO}_3 \downarrow$

C) True statement

D) $\text{CO}_3^{2-} + 2\text{Hg}^{2+} + 3\text{H}_2\text{O} \rightarrow \text{Hg}_4\text{O}_3\text{CO}_3 \downarrow$ (reddish brown) + 6H^+
 (basic mercury (II) carbonate)

9. Ans (A,B,D)



$$nf = 2 \quad nf = 1$$

No. of Meq in Part-I = No. of meq of hypo = 15×0.4

No. of Meq I_2 = 6 meq = 3 millimoles

Part-II



Millimoles of I_2 with NaOH

$$= \frac{(30) - (2 \times 3)}{2} = 12 \text{ millimoles}$$

Total number of millimoles of I_2 = $12 + 3 = 15$ millimoles

$$\text{Initial molarity} = \frac{15 \text{ mmol}}{150 \text{ ml}} = 0.1\text{M}$$

PART-2 : CHEMISTRY

SECTION-III

2. Ans (4)

SO_2 , SO_3 , H_3PO_4 and HClO_4 , has $p\pi-d\pi$ bonding

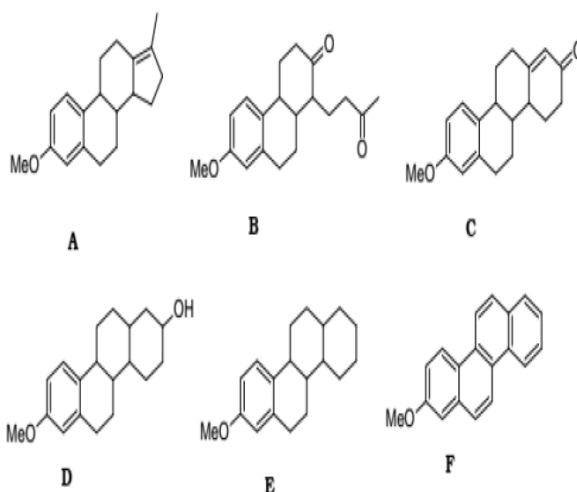
3. Ans (4)

$\text{Hg}^{2+} + \text{Co}(\text{SCN})_2 \rightarrow \text{Co}[\text{Hg}(\text{SCN})_4]$ (deep blue)

4. Ans (8)

A) ClF_3 ; $x = 11$ B) UF_6 ; $y = 18$ C) ClF ; $z = 6$

5. Ans (7)



6. Ans (4)

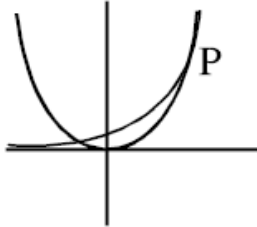
b, c, e, j only

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (D)

Since both curves touches each other



$$\Rightarrow m_1 = m_2$$

$$4(x - a) = 2e^{2x}$$

$$4(x - a) = 2.2(x - a)^2$$

$$\Rightarrow (x - a)(x - a - 1) = 0$$

$$\Rightarrow x = a \text{ or } x = a + 1$$

If $x = a$ then (1st) curve $\Rightarrow y = 0$

& 2nd curve $\Rightarrow y = e^{2a}$ but $e^{2a} \neq 0 \forall a \in \mathbb{R}$

$$\therefore x \neq a$$

$$\Rightarrow x = a + 1$$

$$\therefore 1^{\text{st}} \text{ curve} \Rightarrow y = 2 \therefore (x, y) = (a + 1, 2)$$

(x, y) lies on 2nd curve also, $\therefore 2 = e^{2a+2}$

$$\Rightarrow 2a + 2 = \ln 2$$

$$a = \frac{1}{2}(\ln 2 - 2)$$

$$\Rightarrow -1 < a < 0$$

2. Ans (B)

Let $P(t, t^2)$ be the parametric co-ordinates of any point on the parabola $y = x^2$

$$\Rightarrow \left. \frac{dy}{dt} \right|_{(t, t^2)} = 2t$$

$$\text{Hence normal at P is } (y - t^2) = \frac{-1}{2t}(x - t)$$

If this line meets the curve again at $Q(t', t'^2)$, then

$$(t'^2 - t^2) = \left(-\frac{1}{2t}\right)(t' - t)$$

$$\Rightarrow t' + t = -\frac{1}{2t}$$

$$\Rightarrow t' = -t - \frac{1}{2t}$$

If ℓ is length of the normal chord then $z = (\ell)^2 = PQ^2$

$$\Rightarrow z = (t - t')^2 [1 + (t + t')^2]$$

$$\Rightarrow z = \left(t + t + \frac{1}{2t}\right)^2 \left(1 + \left(-\frac{1}{2t}\right)^2\right)$$

$$\Rightarrow z = \frac{(4t^2 + 1)^3}{16t^4}$$

Put $4t^2 = p$ where $p > 0$

$$z = \frac{(p + 1)^3}{p^2} = (p + 3) + \frac{3}{p} + \frac{1}{p^2}$$

$$\Rightarrow \frac{dz}{dp} = \left(1 - \frac{3}{p^2}\right) - \frac{2}{p^3} = 0$$

$$\Rightarrow (p + 1)^2 (p - 2) = 0$$

$$\Rightarrow p = 0 \text{ as } p \neq -1 \text{ and } \frac{d^2z}{dp^2} = \frac{6}{p^3} + \frac{6}{p^4} > 0$$

Hence z is minimum when $p = 2 = 4t^2$

3. Ans (D)

$$I(k) = \int_0^\infty \frac{\ln x}{x^2 + kx + k^2} dx$$

Put $x = \frac{k}{t}$ we get

$$I(k) = \frac{1}{k} \int_0^\infty \frac{\ln k - \ln t}{t^2 + t + 1} dt$$

$$\Rightarrow I(k) = \frac{\ln k}{k} \int_0^\infty \frac{dt}{t^2 + t + 1} - \frac{I(1)}{k}$$

Now $I(1) = -I(1)$

$$I(k) > \frac{\ln k}{k} \int_0^1 \frac{dt}{t^2 + t + 1} \quad (k > 1)$$

4. **Ans (D)**

Because $|z| \leq 1$ and $|iw| \leq 1$ and $|z - iw| = 2$, so z, iw are end points of diameter of $x^2 + y^2 = 1$.

Similarly z and $i\bar{w}$ are also end points of same diameter (because $|z - i\bar{w}| = 2$)

$\Rightarrow iw = i\bar{w} \Rightarrow iw$ lies on real axis

$\Rightarrow z$ lies on real axis

$\Rightarrow$ either $z = 1, iw = -1$ or $z = -1, iw = 1$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. **Ans (A,C)**

$$\begin{aligned} \text{Let } t_r &= \left(\frac{1}{r(r+1)} \right)^3 = \left(\frac{1}{r} - \frac{1}{r+1} \right)^3 \\ &= \frac{1}{r^3} - \frac{3}{r^2(r+1)} + \frac{3}{r(r+1)^2} - \frac{1}{(r+1)^3} \\ &= \frac{1}{r^3} - \frac{1}{(r+1)^3} - 3 \left(\frac{(r+1) - r}{r(r+1)} \right)^2 \\ t_r &= \frac{1}{r^3} - \frac{1}{(r+1)^3} - 3 \left(\frac{1}{r^2} + \frac{1}{(r+1)^2} \right) + 6 \left(\frac{1}{r} - \frac{1}{r+1} \right) \\ \sum_{r=1}^n t_r &= \lim_{n \rightarrow \infty} \left(1 - \frac{1}{(n+1)^3} \right) - 3 \left(\frac{\pi^2}{6} + \frac{\pi^2}{6} - 1 \right) + 6 \left(1 - \frac{1}{n+1} \right) \\ \lim_{n \rightarrow \infty} \sum t_r &= 10 - \pi^2 \end{aligned}$$

6. **Ans (A,B,C)**

Since $\sec^2 \theta > 1$

$$\Rightarrow [(n+1) \sec^2 \theta] > [n \sec^2 \theta]$$

Hence, f and g are both one-one

Let $k < n \sec^2 \theta < k+1$ and

$k < m \operatorname{cosec}^2 \theta < k+1$

$k \cos^2 \theta < n < (k+1) \cos^2 \theta$ and

$k \sin^2 \theta < m < (k+1) \sin^2 \theta$

Adding given $k < \text{integer} < k+1$

$$\Rightarrow A \cap B = \phi$$

Suppose $k \notin A$

$$\Rightarrow n \sec^2 \theta < k \text{ and } (k+1) < (n+1) \sec^2 \theta$$

$$\Rightarrow n < k \cos^2 \theta \text{ and } n+1 > (k+1) \cos^2 \theta$$

$$\Rightarrow k < (k-n) \operatorname{cosec}^2 \theta, k+1 > (k-n) \operatorname{cosec}^2 \theta$$

$$\Rightarrow [(k-n) \operatorname{cosec}^2 \theta] = k$$

$$\Rightarrow k \in B$$

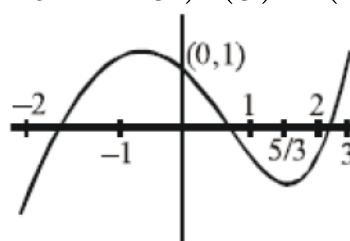
7. **Ans (B,C)**

Required probability =

$$\begin{aligned} & \frac{P(P_1 \text{ reaches final} \cap P_2 \text{ reaches semifinal but not in final})}{P(P_2 \text{ reaches semifinal but not in final})} \\ &= \frac{P(P_1 \text{ and } P_2 \text{ reaches semifinal} \cap P_1 \text{ wins semifinal but } P_2 \text{ loses semifinal})}{P(P_2 \text{ reaches semifinal and } P_2 \text{ loses semifinal})} \\ &= \frac{\frac{{}^{14}C_2}{{}^{16}C_4} \times \left(\frac{1}{3}\right) \left(\frac{1}{2} + \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2}\right)}{\frac{{}^{15}C_3}{{}^{16}C_4} \times \frac{1}{2}} = \frac{2}{15} \end{aligned}$$

8. **Ans (B,D)**

$$= \left(\frac{14 \times 13 \times 12}{16 \times 15 \times 14 \times 13} \right) \left(\frac{8}{3} \right) = \left(\frac{2}{15} \right)$$



$$\text{Clearly, } \begin{vmatrix} x & x & 5 \\ 1 & x & 0 \\ 1 & 1/5 & x \end{vmatrix} = 0 \Rightarrow x^3 - x^2 - 5x + 1 = 0$$

$$\text{Assume } f(x) = x^3 - x^2 - 5x + 1$$

$$\therefore f'(x) = 3x^2 - 2x - 5 = 0 \Rightarrow x = -1 \text{ \& } \frac{5}{3}$$

$$\text{Clearly } [\alpha] + [\beta] + [\gamma] = -2 + 0 + 2 = 0$$

$$\text{Now } x^3 - x^2 - 5x + 1 = (x - \alpha)(x - \beta)(x - \gamma)$$

Put $x = -1$, we get

$$(1 + \alpha)(1 + \beta)(1 + \gamma)$$

$$= -(-1 - 1 + 5 + 1) = -4$$

Now, volume of parallelepiped

$$= |x^3 - x^2 - 5x + 1| \times \in [-2, 3]$$

Maximum value will occur at $x = \frac{5}{3}$ and is equal to

$$\left| \frac{125}{27} - \frac{25}{9} - \frac{25}{3} + 1 \right| = \left| \frac{125 - 75 - 225 + 27}{27} \right| = \frac{148}{27}$$

9. **Ans (B,C,D)**

Orthocentre of ΔPQR lies on hyperbola

$$\Rightarrow \lambda = 3$$

Incentre of ΔABC is orthocenter of ΔPQR

$$\text{Equation of AB is } x + y - \frac{26}{5} = 0$$

10. Ans (A,C,D)

Obviously $f(1 + \alpha) = f(1 - \alpha) \forall \alpha \in (0,1)$

Domain of f is $(0, 2)$

Hence (A) is true

Maxima of $\ln(2x - x^2)$ as well as that of

$\sin x \left(\frac{\pi x}{2} \right)$ occur at $x = 1$.

Hence (C) is true

As $x \rightarrow 0^+$ or $x \rightarrow 2^-$, $f(x) \rightarrow -\infty$

Hence (D) is true.

PART-3 : MATHEMATICS

SECTION-III

1. Ans (3)

$$f(2) = 8 - (r-1)4 - 22 + 4r = -ve$$

$$f(-2) = -8 - (r-1)4 + 22 + 4r = +ve$$

$\Rightarrow$ One of the integral root can be, -1 or 0 or 1

Case - I : $x = -1$ is one root $\Rightarrow r = -\frac{11}{3}$ rejected

Case - II : $x = 0$ is one root $\Rightarrow r = 0$ but for $r = 0$, $f(x) = x^3 - x^2 - 11x$ which does not have all integral roots

$\Rightarrow r = 0$ (rejected)

Case - III : $x = 1$ is one root $\Rightarrow r = 3$ and at $r = 3$ all roots are integral

$$(\text{Equation is } x^3 - 2x^2 - 11x + 12 = 0)$$

$$\Rightarrow (x-1)(x-4)(x+3) = 0$$

2. Ans (4)

$$4I + 3J = \int_0^{\frac{\pi}{2}} 1 dx = \frac{\pi}{2}$$

$$3I - 4J = \int_0^{\frac{\pi}{2}} \frac{3 \cos x - 4 \sin x}{3 \sin x + 4 \cos x + 25} dx$$

$$= [\ln(3 \sin x + 4 \cos x + 25)]_0^{\frac{\pi}{2}} = \ln 28 - \ln 29$$

$$3I - 4J = \ln \left(\frac{28}{29} \right) \quad \dots(i)$$

$$4I + 3J = \frac{\pi}{2} \quad \dots(ii)$$

$$11I + 2J = \pi + \ln \left(\frac{28}{29} \right) \equiv 2\pi + b \ln \left(\frac{c}{d} \right)$$

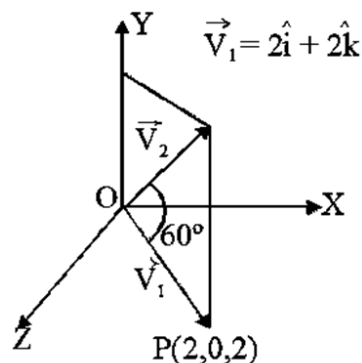
$$\Rightarrow d = 1 \text{ and } b = 1, c = 28, d = 29$$

$$\therefore a + b + c + d - 55 = 59 - 55 = 4$$

3. Ans (4)

Equation of plane perpendicular to XOZ plane is

$P : x - z = 0$, $\vec{V}_2$ is coplanar with $\vec{V}_1$ and $\hat{j}$.



$$\Rightarrow \vec{V}_2 = x\vec{V}_1 + y\hat{j}$$

$$\Rightarrow \vec{V}_2 = x(2\hat{i} + 2\hat{k}) + y\hat{j}$$

$$\therefore |\vec{V}_2| = |\vec{V}_1| = 2\sqrt{2}$$

$$\therefore 4x^2 + y^2 + 4x^2 = 8 \quad \dots(i)$$

$$\text{Now, } \vec{V}_2 \cdot \vec{V}_1 = |\vec{V}_2| |\vec{V}_1| \cos \frac{\pi}{3} = 4$$

$$4 = (x(2\hat{i} + 2\hat{k}) + y\hat{j}) \cdot (2\hat{i} + 2\hat{k})$$

$$4 = 8x \Rightarrow x = \frac{1}{2} \Rightarrow y = \pm\sqrt{6}$$

$$\Rightarrow \vec{V}_2 = \hat{i} + \sqrt{6}\hat{j} + \hat{k} \quad (\because \vec{V}_2 \cdot \hat{j} > 0)$$

$$\text{Now, } \vec{V}_3 = \vec{V}_1 + x\hat{i}$$

$$\vec{V}_3 \cdot \vec{V}_1 = 0 \Rightarrow 8 + 2x = 0 \Rightarrow x = -4$$

$$\Rightarrow \vec{V}_3 = -2\hat{i} + 2\hat{k}$$

$\Rightarrow$ Volume of tetrahedron

$$V = \left| \frac{1}{6} \begin{vmatrix} 2 & 0 & 2 \\ 1 & \sqrt{6} & 1 \\ -2 & 0 & 2 \end{vmatrix} \right|$$

$$\Rightarrow V = \frac{1}{6} 8\sqrt{6} \Rightarrow \frac{3V}{\sqrt{6}} = 4$$

4. **Ans (3)**

$$h'(x) = f(x)$$

$$h''(x) = f'(x)$$

$$h(1) = 0, f(1) = f'(1) = h'(1) = h''(1) = 1 = g(1)$$

$$\Rightarrow f'(g(x)) g'(x) = 1$$

$$\Rightarrow f'(g(1))g'(1) = 1 \Rightarrow f'(1)g'(1) = 1 \Rightarrow g'(1) = 1$$

$$G(x) = x^2 g(x) - x h(g(x))$$

$$G'(x) = 2xg(x) + x^2 g'(x) - h(g(x)) - xh'(g(x))g'(x)$$

$$= 2xg(x) + x^2 g'(x) - h(g(x)) - x^2 g'(x) = 2xg(x) - h(g(x))$$

$$G''(x) = 2g(x) + 2xg'(x) - h'(g(x))g'(x) = 2g(x) + xg'(x)$$

$$\therefore G''(1) = 2g(1) + g'(1) = 2 + 1 = 3$$

5. **Ans (0)**

$$(x+1)^3 + (y+1)^3 = 16$$

$$y = -1 + \left(16 - (x+1)^3\right)^{\frac{1}{3}} = f(x)$$

$$x = -1 + \left(16 - (y+1)^3\right)^{\frac{1}{3}}$$

$$\therefore f^{-1}(x) = -1 + \left(16 - (x+1)^3\right)^{\frac{1}{3}} = g(x)$$

$$\Rightarrow f(x) = g(x)$$

$$f(-1 + \sqrt[3]{15}) = g(-1 + \sqrt[3]{15}) = 0$$

$$(f(x+g(x)).g(x+f(x))) = (f(x+f(x)))^2$$

$$\frac{d}{dx}(f(x+f(x)))^2 = 2f(x+f(x))$$

$$f'(x+f(x)).(1+f'(x))$$

$$\frac{d}{dx}(f(x+f(x)))^2 \text{ at } x = -1 + \sqrt[3]{15} = 0$$

$$\therefore g(-1 + \sqrt[3]{15}) = f(-1 + \sqrt[3]{15}) = 0$$

6. **Ans (1)**

$$P_1 : x + y - z = 1$$

$$P_2 : (m-1)x + (3m-7)y - 5z = -3$$

$$P_3 : (m-2)x + (2m-5)y = -4$$

$\therefore$ all the three planes intersects at a line

$$\therefore \begin{vmatrix} 1 & 1 & -1 \\ m-1 & 3m-7 & -5 \\ m-2 & 2m-5 & 0 \end{vmatrix}$$

$$\Rightarrow m = 3, -2$$

At $m = 3$

$$\left. \begin{array}{l} P_1 : x + y - z = 1 \\ P_2 : 2x + 2y - 5z = -3 \\ P_3 : x + y = -4 \end{array} \right\} \text{System does not}$$

have any solution

but at $m = -2$

$$P_1 : x + y - z = 1$$

$$P_2 : 3x + 13y + 5z = 3$$

$$P_3 : 4x + 9y = 4$$

$$\Rightarrow 5P_1 + P_2 = 2P_3$$

$\Rightarrow$ Direction ratio of line of intersection are given by

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -1 \\ 3 & 13 & 5 \end{vmatrix} = 18\hat{i} - 8\hat{j} + 10\hat{k}$$

$$\equiv \langle 9, -4, 5 \rangle$$

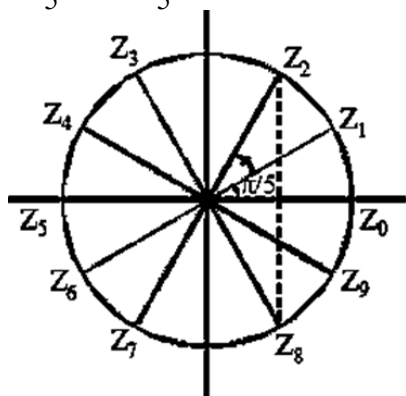
$$\Rightarrow a = 9\lambda, b = -4\lambda, c = 5\lambda$$

$$\therefore \frac{a+b}{c} = 1$$

7. Ans (8)

$$z = \cos \frac{\pi}{5} + 2i \sin \frac{\pi}{10} \cos \frac{\pi}{10}$$

$$\Rightarrow z = \cos \frac{\pi}{5} + i \sin \frac{\pi}{5} = e^{i\frac{\pi}{5}}$$



$$\because Z_p = z^p = e^{ip\frac{\pi}{5}}, p = 0, 1, 2, 3, \dots, 9$$

$$(z_2, z_0)(z_3, z_1)(z_4, z_2)(z_5, z_3)(z_6, z_4)(z_7, z_5)(z_8, z_6)(z_9, z_7)$$

$\Rightarrow$ 8 ordered pairs

8. Ans (7)

$$\frac{dy}{dx} = (x+1) + \frac{y-3}{x+1}$$

$$y = x^2 + x + 3 + c(x+1);$$

$$\because y_{(x=2)} = 0 \text{ so } C = -3$$

$$f(x) = x^2 - 2x$$

$$\text{Area} = \int_0^2 (2x - x^2) dx = \frac{4}{3}; m + n = 7$$

